MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

OVERALL SEVERITY: CRITICAL

Case ID	`case01_baseline_vs_case03_t1059_attack`
Generated	`2026-03-12 00:20:12 UTC`
Baseline capture	`case01_baseline`
Attack capture	`case03_t1059_attack`
Baseline path	`/MFF/cases/case01_baseline`
Attack path	`/MFF/cases/case03_t1059_attack`

1. EXECUTIVE SUMMARY

Metric	Value	Metric	Value
New Processes (Attack Only)	16	Critical Risk Processes	1
Gone Processes (Baseline Only)	16	ATT&CK Techniques Observed	1
New Network Connections	133	ATT&CK Tactics Observed	1
Flagged Connections	4	IOCs Extracted	56

ATT&CK; Tactics Observed:

Execution

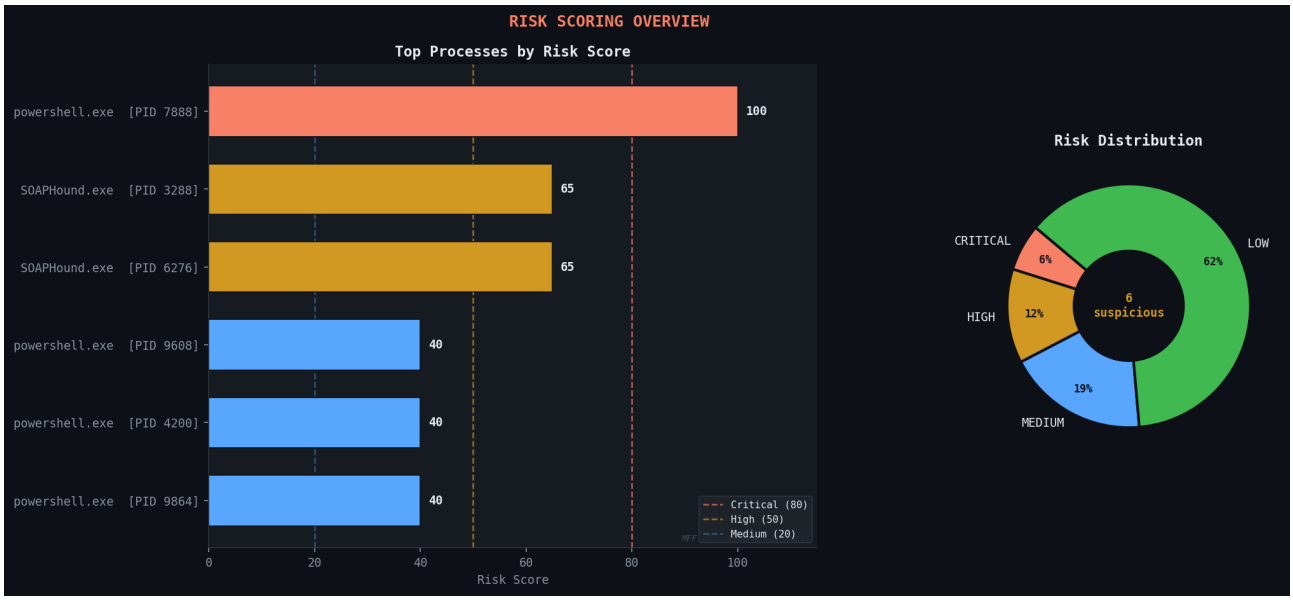
IOC Summary:

Type	Indicators (top 8)
IPv4	0.0.0.0 · 13.107.213.40 · 13.89.179.9 · 172.183.7.192 · 172.183.7.194 · 184.31.68.248

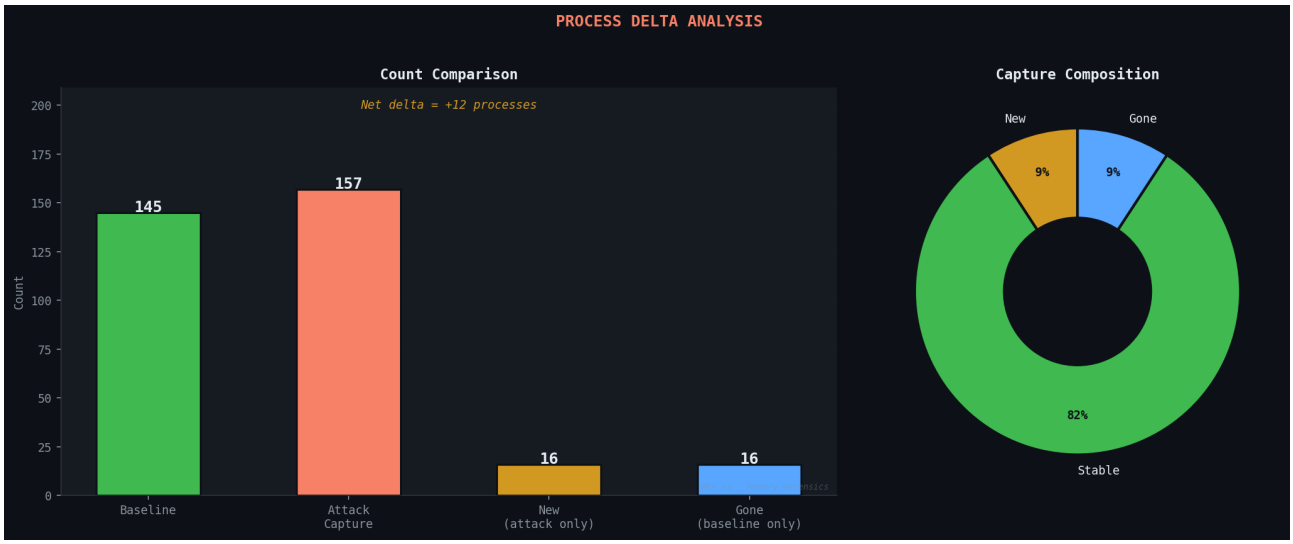
2. VISUALIZATIONS

2.1 Full Forensics Dashboard

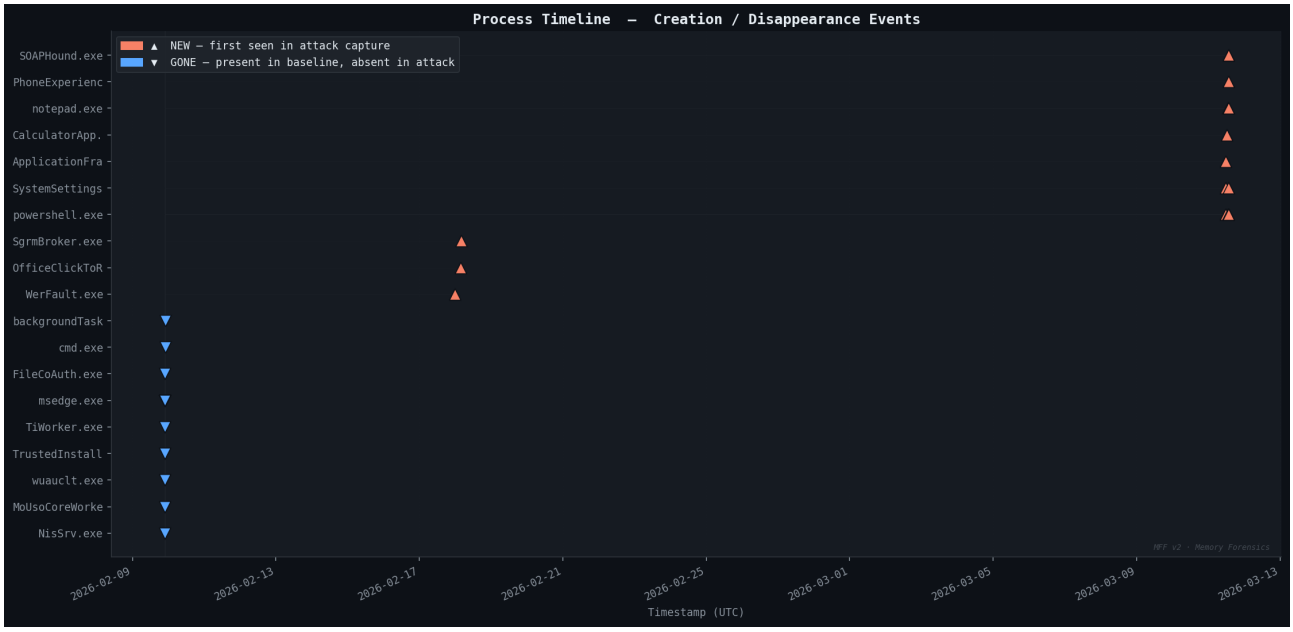
2.2 Risk Scoring Overview



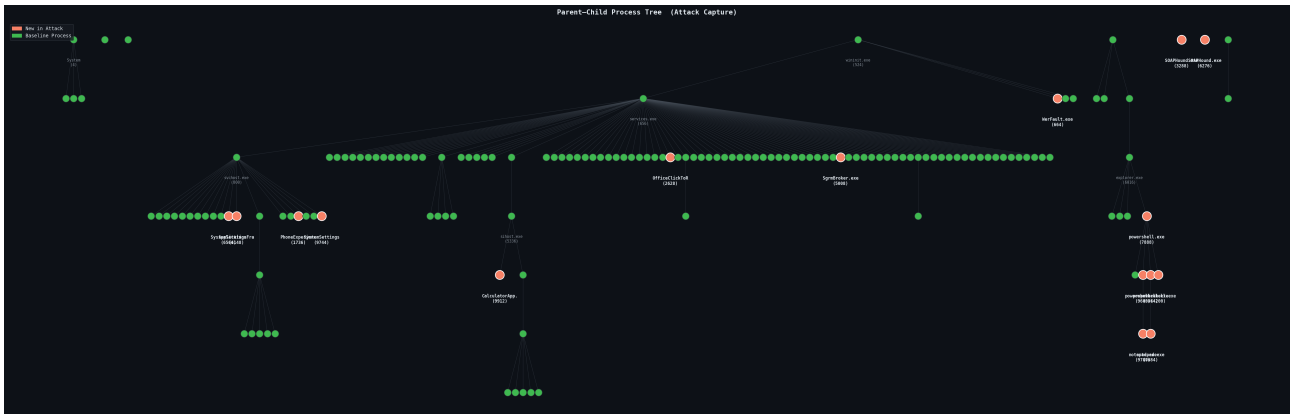
2.3 Process Delta Analysis



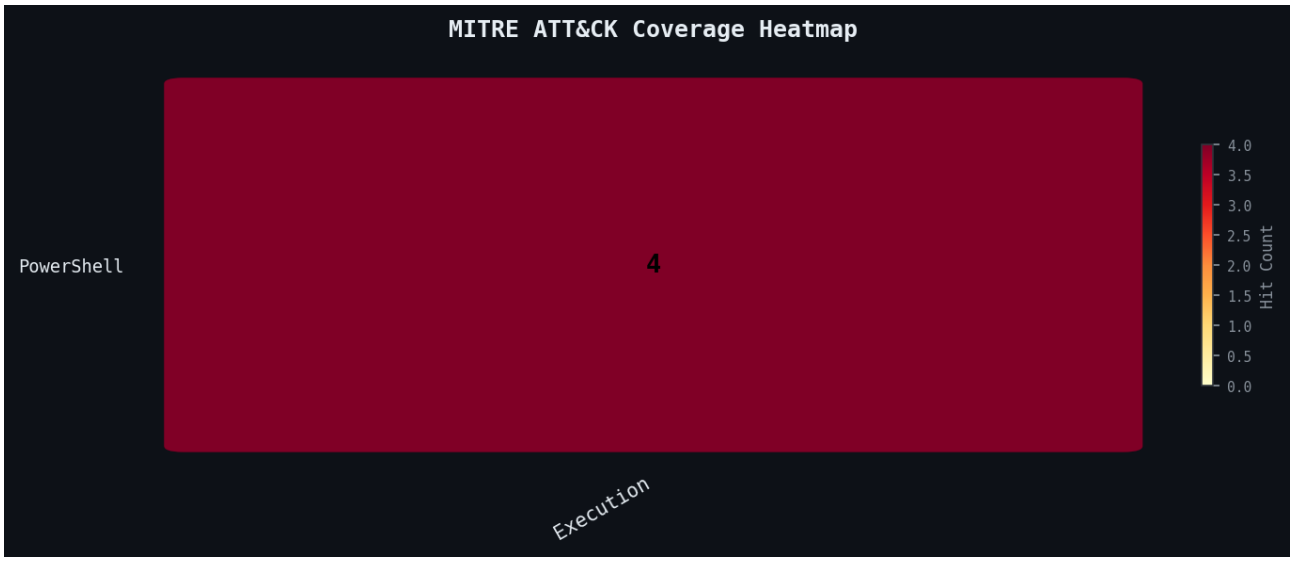
2.4 Process Timeline



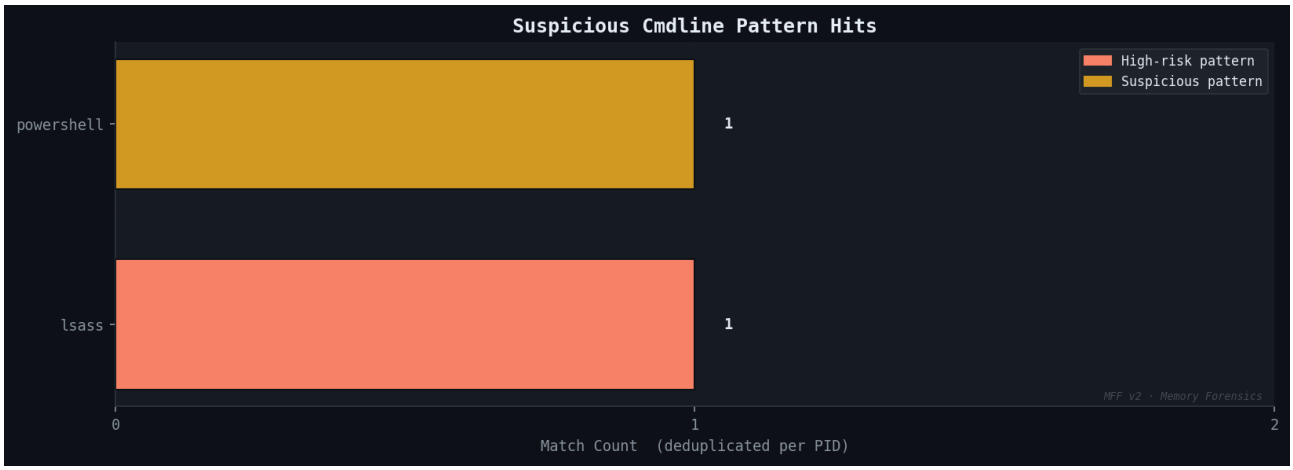
2.5 Parent-Child Process Tree


Parent-Child Process Tree (Attack Capture)
New in Attack
Baseline Process
System
(4)
wininit.exe
(544)
services.exe
(684)
svchost.exe
(800)
SgrmBroker.exe
(5408)
MsMpEng.exe
(4016)
powershell.exe
(7988)
CalculatorApp.
(8922)
SDXHelperBackgroundHost.exe
(5208)
(8176)
WerFault.exe
(964)
OfficeClickToR
(2628)
(5104)
(1216)
(9704)
(5016)
(7880)
(7880)
(7880

2.6 MITRE ATT&CK; Heatmap



2.7 Suspicious Cmdline Patterns



2.8 Malfind Memory Protections

3. RISK SCORE TABLE

Process	PID	Score	Level	Indicators
powershell.exe	7888	100	CRITICAL	Suspicious cmdline pattern; RWX executable memory region (malfind)
SOAPHound.exe	3288	65	HIGH	Known-malicious name: SOAPHound.exe; Anomalous process name keyword
SOAPHound.exe	6276	65	HIGH	Known-malicious name: SOAPHound.exe; Anomalous process name keyword
powershell.exe	9608	40	MEDIUM	Suspicious cmdline pattern
powershell.exe	4200	40	MEDIUM	Suspicious cmdline pattern
powershell.exe	9864	40	MEDIUM	Suspicious cmdline pattern

4. NEW PROCESSES (Attack Only)

ImageFileName	PID	PPID	CreateTime
WerFault.exe	664	524	2026-02-17 23:49:52.000000 UTC
OfficeClickToR	2628	656	2026-02-18 03:45:58.000000 UTC
SgrmBroker.exe	5008	656	2026-02-18 03:48:05.000000 UTC
powershell.exe	7888	6016	2026-03-11 11:48:11.000000 UTC
SystemSettings	6564	800	2026-03-11 11:50:00.000000 UTC
ApplicationFra	6148	800	2026-03-11 11:50:18.000000 UTC
CalculatorApp.	9912	5336	2026-03-11 12:18:39.000000 UTC
powershell.exe	9608	7888	2026-03-11 13:18:21.000000 UTC
notepad.exe	9780	9608	2026-03-11 13:18:21.000000 UTC
powershell.exe	9864	7888	2026-03-11 13:24:48.000000 UTC
notepad.exe	7684	9864	2026-03-11 13:24:49.000000 UTC
PhoneExperienc	1736	800	2026-03-11 13:28:34.000000 UTC
powershell.exe	4200	7888	2026-03-11 13:29:21.000000 UTC
SOAPHound.exe	6276	9816	2026-03-11 13:32:02.000000 UTC
SOAPHound.exe	3288	9508	2026-03-11 13:32:03.000000 UTC
SystemSettings	9744	800	2026-03-11 13:32:54.000000 UTC

5. MITRE ATT&CK; TECHNIQUE MATCHES

Technique	TechniqueName	Tactic	MatchedKeyword	PID	Process
T1059.001	PowerShell	Execution	powershell	7888	powershell.exe
T1059.001	PowerShell	Execution	powershell	9608	powershell.exe
T1059.001	PowerShell	Execution	powershell	9864	powershell.exe
T1059.001	PowerShell	Execution	powershell	4200	powershell.exe

6. SUSPICIOUS CMDLINE FINDINGS

PID	Matched Pattern	Args
7888	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
688	lsass	C:\\Windows\\system32\\lsass.exe

7. MALFIND — MEMORY INJECTION ARTEFACTS

Process	PID	Protection
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE

8. NETWORK ARTEFACTS

8a. Flagged Connections (suspicious ports)

TreeDepth	Offset	Proto	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created	SuspiciousPort	PortMeaning
0	0xa4827a562e10	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	920	svchost.exe	2026-02-17 23:49:54.000000 UTC	135	DCOM/RPC
0	0xa4827a563230	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	920	svchost.exe	2026-02-17 23:49:54.000000 UTC	135	DCOM/RPC
0	0xa4827a563230	TCPv6	::	135	::	0	LISTENING	920	svchost.exe	2026-02-17 23:49:54.000000 UTC	135	DCOM/RPC
0	0xa4827f8f3890	TCPv4	10.0.2.5	139	0.0.0.0	0	LISTENING	4	System	2026-03-11 13:27:02.000000 UTC	139	NetBIOS

9. IOC DETAIL TABLE

Type	Value	Source	Count
FilePath	C:\\Windows\\system32\\svchost.exe	cmdline	49
FilePath	C:\\Windows\\System32\\svchost.exe	cmdline	28
FilePath	C:\\Program	cmdline	12
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView	cmdline	7
FilePath	Microsoft\\EdgeWebView\\Application\\144.0.3719.115\\msedgewebview2.exe	cmdline	7
FilePath	C:\\Windows\\System32\\RuntimeBroker.exe	cmdline	6
FilePath	.....	malfind	4
FilePath	C:\\ProgramData\\Microsoft\\Windows	cmdline	2
FilePath	C:\\Windows\\system32\\DllHost.exe	cmdline	2
FilePath	Microsoft	cmdline	2
FilePath	Windows	cmdline	2

Type	Value	Source	Count
FilePath	system32\\csrss.exe	cmdline	2
FilePath	.....	malfind	1
FilePath	.....	malfind	1
FilePath	.....	malfind	1
FilePath	??\\C:\\Windows\\system32\\conhost.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Microsoft\\EdgeUpdate\\MicrosoftEdgeUpdate.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView\\Crashpad	cmdline	1
FilePath	C:\\Windows\\Explorer.EXE	cmdline	1
FilePath	C:\\Windows\\ImmersiveControlPanel\\SystemSettings.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SecurityHealthSystray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SystemSettingsBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxService.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxTray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	1
FilePath	C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\spoolsv.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.Search_cw5n1h2txyewy\\SearchApp.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\\TextInputHost.exe	cmdline	1

10. METHODOLOGY

Dataset: Memory images captured using VMware hypervisor snapshot of Windows 10 VM. Attack simulations performed with Atomic Red Team (ART) scripts.

Volatility 3: Memory parsed using Volatility 3 plugins: pslist, cmdline, malfind, netscan, pstree. Output exported as CSV.

Process diff: Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs reassigned on every reboot.

Risk scoring: Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline +40 · Known-malicious name +50 · Anomaly bonus +15.

Malfind filter: Known-clean JIT processes excluded (browsers, Defender, .NET). Only genuinely suspicious RWX regions retained.

ATT&CK; mapping: 25+ rule-based signatures across 8 tactics. Keyword matching on cmdline args, process names, and network artefacts.

IOC extraction: Regex-based extraction: IPv4 (excluding RFC1918), domains, URLs, MD5/SHA1/SHA256, Windows file paths.

Framework: MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project. Tool chain: Volatility 3 → MFF v2 → HTML/PDF reports.